

Project Title: Quantum Framework for Solving Nonlinear Partial Differential Equations in Physical Systems

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Project Overview: Many real-world physical systems are governed by nonlinear differential equations, which are often computationally expensive to solve using classical methods. This project aims to explore a hybrid classical–quantum framework for solving such systems by combining linearization techniques with quantum algorithms. The approach focuses on transforming nonlinear systems into linear representations and solving them using quantum algorithms such as Variational Quantum Linear Solver, Harrow-Hassidim-Lloyd Algorithm, and Quantum Neural Networks (QNNs). The results will be benchmarked against classical numerical methods to evaluate performance, scalability, and accuracy. Interns will gain experience in quantum computing, numerical methods, hybrid modeling, and scientific research practices.

Objectives

- Develop a general framework for solving nonlinear differential equations using hybrid quantum-classical methods
- Apply linearization techniques to transform nonlinear systems into solvable linear forms
- Implement quantum solvers (VQLS, HHL, QNN-based approaches)
- Compare results with classical numerical solvers (FDM, FEM, PINNs, RBF)
- Analyze performance in terms of accuracy, scalability, and computational complexity
- Train interns in scientific computing, quantum programming, and research methodology

Weekly Plan

Week 1: Orientation & Foundations

- Introduction to nonlinear physical systems and differential equations
- Overview of linearization concepts
- Introduction to quantum computing tools (Qiskit, PennyLane)
- Literature review on quantum linear solvers
- **Deliverables:** Literature summary + selected problem class

Week 2: Problem Definition & Preprocessing

- Selection of representative nonlinear problems
- Mathematical formulation of systems
- Application of general linearization techniques
- Discretization and preparation of linear systems
- **Deliverables:** Linearized system formulation + technical note

Week 3: Quantum Model Design

- Mapping linear systems into quantum representations

- Design of circuits for VQLS, HHL, and QNN approaches
- Setup of simulation environments
- **Deliverables:** Quantum model design + circuit architecture

Week 4: Quantum Implementation

- Implementation of quantum solvers using simulators
- Initial testing and debugging
- Evaluation of convergence behavior
- **Deliverables:** Working quantum solver + initial results

Week 5: Analysis & Paper Drafting

- Study of solver behavior (stability, convergence trends)
- Resource analysis (qubits, circuit depth)
- Drafting the research paper:
 - Introduction
 - Methodology
 - Preliminary results
- **Deliverables:** Analysis report + first draft of paper

Week 6: Finalization, Paper & Presentation

- Refinement of quantum models and experiments
- Finalization of results and visualizations
- Completion of research paper draft
- Preparation of final presentation
- Discussion of publication opportunities
- **Deliverables:** Final report + paper draft + presentation + code repository

Expected Outcomes

- A quantum-based framework for solving linearized nonlinear systems
- Implementation of multiple quantum solvers
- Analysis of convergence and computational characteristics
- A draft research paper suitable for submission
- Reproducible code and documentation